

RVHS JC2 H2 Physics Preliminary Examinations Paper 3 Mark Scheme

- 1 (a) Systematic error: forgetting to subtract the original length of the spring B1
Random error: parallax error when reading the extension of the spring B1
- (b) (i) Any other logical answers.
 $W = F \cos \theta$
 $W = (0.60)(0.750)(\cos 30) = 0.3897 \text{ J}$ C1
 $\text{Max } W = (0.61)(0.751)(\cos 29) = 0.4007 \text{ J}$
 $\text{Min } W = (0.59)(0.749)(\cos 31) = 0.3788 \text{ J}$
Uncertainty of $W = 0.0110 = 0.01$ C1
- Percentage uncertainty = $0.01 / 0.39 * 100\% = 2.6 \%$ A1

2 (a)